Rishi Hazra

Postdoctoral Researcher | FAIR (Meta Superintelligence Labs), London

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RESEARCH & PROFESSIONAL EXPERIENCE

Postdoctoral Researcher Jul 2025 – Present
FAIR (Meta Superintelligence Labs), London, UK
AI Research Agents.

Research Scientist Intern Aug 2024 – Feb 2025
FAIR (Meta AI), London, UK
SAM 3; AI Research Agents.

Research Scientist Intern Jul 2022 – Dec 2022
Meta Reality Labs Research, Redmond, USA
Vision and language-based task tracking.

Data Scientist Apr 2020 – Sep 2020
Amazon Alexa-AI, Bangalore, India
NLU metrics for Alexa.

Research Associate Jun 2019 – Mar 2020
Statistics & Machine Learning Group, IISc Bangalore
Multi-agent reinforcement learning.

EDUCATION

Ph.D. 2021 – 2025
Örebro University & WASP, Sweden
Thesis: Neurosymbolic decision-making with LLMs.
Supervisors: Luc De Raedt, Pedro Zuidberg Dos Martires.

M.Tech in Artificial Intelligence 2017 – 2019
Indian Institute of Science, Bangalore
Thesis: Active learning in sequence tagging.
GPA: 8.10/10. Supervisor: Ambedkar Dukkipati.

B.Tech in Electrical Engineering 2013 – 2017
Birsa Institute of Technology, India
GPA: 8.03/10.

SELECTED PUBLICATIONS

1. **AIRA$_2$: Overcoming Bottlenecks in AI Research Agents**
   K Hambardzumyan*, N Baldwin*, E Toledo*, R Hazra*, M Kuchnik*, M Josifosky* et al. (*: equal contribution)
   NeurIPS 2026 [pdf]
2. **AI Research Preference Models**
   Thomas Simon Foster*, Bassel Al Omari*, Tingchen Fu*, Thomas Mann, Carl Domond, et al.
   Preprint 2026 [pdf]
3. **Training AI Co-Scientists Using Rubric Rewards**
   S Goel, R Hazra, D Jayalath, T Willi, P Jain, WF Shen, I Leontiadis, F Barbieri, Y Bachrach, J Geiping, C Whitehouse
   ICML 2026 [pdf] [dataset]
4. **COvolve: Adversarial Co-Evolution of Large-Language-Model-Generated Policies and Environments via Two-Player Zero-Sum Game**
   A Sygkounas, R Hazra, A Persson, PZD Martires, A Loutfi
   GECCO 2026 [pdf]
5. **SAM 3: Segment Anything with Concepts**
   N Carion et al.
   ICLR 2026 [pdf] [code] [website]
6. **AI Research Agents for ML: Search, Exploration, and Generalization in MLE-bench**
   E Toledo, K Hambardzumyan, M Josifoski, R Hazra et al.

NeurIPS 2025 (Spotlight) [pdf] [code]

7. **LexiCon: Planning under Temporal Constraints in Natural Language**
   P Mantenoglou, R Hazra, PZD Martires, L De Raedt
   NeurIPS 2025 (DB Track) [pdf] [code]
8. **Have Large Language Models Learned to Reason? A Characterization via 3-SAT**
   R Hazra, G Venturato, PZD Martires, L De Raedt
   COLM 2025 [pdf] [code]
9. **REvolve: Reward Evolution with Large Language Models using Human Feedback**
   R Hazra*, A Sygkounas*, A Persson, A Loutfi, PZD Martires
   ICLR 2025 [web] [pdf] [code]
10. **SayCanPay: Heuristic Planning with LLMs using Learnable Domain Knowledge**
    R Hazra, PZD Martires, L De Raedt
    AAAI 2024 [web] [pdf] [code]
11. **EgoTV: Egocentric Task Verification from Natural Language Task Descriptions**
    R Hazra, B Chen, A Rai, N Kamra, R Desai
    ICCV 2023 [web] [pdf] [code]
12. **Deep Explainable Relational Reinforcement Learning: A Neuro-Symbolic Approach**
    R Hazra, L De Raedt
    ECML PKDD 2023 [pdf]
13. **Active^2 Learning: Reducing Redundancies in Active Learning for Sequence Tagging**
    R Hazra, P Dutta, S Gupta, MA Qaathir, A Dukkipati
    NAACL-HLT 2021 [pdf] [code]
14. **Networked Multi-Agent Reinforcement Learning with Emergent Communication**
    S Gupta*, R Hazra*, A Dukkipati
    AAMAS 2020 [pdf]

Workshop papers (ViGIL @ NAACL-HLT 2021, IEEE RO-MAN 2024, HRI 2025 LBR) omitted for brevity.

MENTORSHIP

Master Thesis Supervision 2024 – 2025
Jens V Rüppel, TU Chemnitz (Co-supervisor: Tim Schreiter)

ACADEMIC SERVICE

Program Committee & Session Chair: AAMAS 2022

Reviewer: NeurIPS 2022 (Top), 2023, 2024 (Top), 2025, 2026 | ICML 2023–25 | ICLR 2024–25 | KR 2024 | EACL 2023 | TMLR

COMMUNITY

- **PRAYAAS India** (2013–2016): Taught mathematics to underprivileged children.
- **Tarumitra** (2011–2013): Student President; led plantation drives and awareness programs.